

Fine-root morphological traits regulate soil infiltration rate through soil properties in arid shrublands

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ABSTRACT

Vegetation restoration plays a vital role in influencing soil hydrological processes in arid regions. Root-soil interactions alter soil water infiltration, which governs the conversion of precipitation into soil water. However, how fine-root morphological traits regulate soil infiltration by mediating the change of soil structure and hydraulic properties remains unclear. To address this, fine-root (diameter ≤ 2 mm) morphological traits (specific root length (SRL), specific root surface area (RSA), root tissue density (RTD), root diameter (RD), and root length density (RLD)) were measured in three typical shrub species (*Caragana sinica*, *Artemisia sphaerocephala*, *Zygo-phyllosum xanthoxylum*) and one grass species (*Astragalus laxmannii*) at 20–40 cm soil depth in a sandy loam soil. Soil physical properties were measured at 0–20 cm soil depth. Structural equation model was used to analyze the effect of fine-root morphological traits on soil infiltration through soil structure and hydraulic characteristic. The result indicated that RTD was a key trait constraining soil infiltration, as it increased capillary porosity and field capacity, thereby enhancing soil water holding capacity and significantly reducing the infiltration rate. Conversely, larger RD and RSA facilitated the formation of more stable and interconnected root channels, promoting water infiltration. The steady infiltration rate of *A. sphaerocephala* ranked second only to *C. sinica*, yet was 48% and 63% higher than that of *A. laxmannii* and *Z. xanthoxylum*, respectively. *A. sphaerocephala* also exhibit higher RSA and SRL, which enabled relatively well infiltration performance while maintaining higher soil aggregate stability. Moreover, *A. laxmannii* and *Z. xanthoxylum* showed higher water retention at field capacity, while *A. sphaerocephala* exhibited higher infiltration rates. These differences suggest potential complementarity in surface soil hydrological functions, but direct evidence for synergistic effects under field conditions is lacking.

1. Introduction

Vegetation restoration is one of the most effective measures to improve soil quality, prevent water and wind erosion, reduce or even reverse the land desertification in arid land (Lal, 2004; Wang et al., 2007). Soil water availability is the primary factor restricting vegetation persistence and functional stability in water limited regions, while vegetation restoration further alters the spatiotemporal pattern of water distribution in the soil-plant system. On the one hand, vegetation modifies rainfall redistribution through canopy interception, stemflow and canopy evaporation. On the other hand, plant-soil interactions modify soil properties, such as soil pore structure, aggregate stability, water-holding capacity and infiltration, thereby regulating the soil water balance (Stavi et al., 2009). In desertified areas, plants such as

Alhagi sparsifolia and *Karelinia caspia* often develop deep root systems to access groundwater or deep soil water, enabling their persistence under water-limited conditions (Tariq et al., 2025). Soil infiltration is a critical hydrological function that governs the conversion of rainfall to soil water and determines water infiltration depth (Wang et al., 2012). Thus, it directly controls the amount of water recharge to deep soil layers and, for deep-rooted plants, the availability of soil water for plant uptake. Soil infiltration is inherently influenced by soil structure and water-holding capacity, which play vital roles in water movement and retention (Bronick and Lal, 2005; Bienes et al., 2016; Huang et al., 2017).

Vegetation cover has been recognized as a key driver of variation in soil infiltration (Sun et al., 2018). Positive effects of vegetation on soil infiltration have been widely reported across multiple ecosystems, including arid regions (Thompson et al., 2010; Rong et al., 2025; Duan

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et al., 2026). During restoration, roots development directly and indirectly modulates soil infiltration by forming root networks and altering soil structure and hydraulic properties through root decay (Shi et al., 2021). Previous studies have indicated that the effects of plant roots on soil infiltration are strongly mediated by interactions between roots and soil properties, including bulk density, porosity and aggregate stability (Wu et al., 2016; Huang et al., 2017). Root growth and penetration enlarge soil microporosity, enhancing soil infiltration rate (Stavi et al., 2009). Additionally, root decay induces the formation of root channels, which further increase soil macropores and facilitate water infiltration (Guo et al., 2019). Root exudates also influence soil aggregate stability and soil water repellency, which in turn affect water infiltration, but the net effect depends on exudate properties and soil conditions (Shi et al., 2021; Zhai et al., 2025). Peral et al. (2026) showed that in dry soil, root architecture and root exudation act synergistically to increase soil hydraulic conductivity, with dissolved root exudates having a significant effect on water infiltration even in the absence of roots. Notably, roots of different diameter exert distinct effects on soil properties, including soil bulk density, porosity, aggregates stability and water-holding capacity (Bardgett et al., 2014; Guo et al., 2019). On the one hand, larger roots may push soil particles aside that increase soil bulk density, whereas fine roots can reduce bulk density by enhancing soil porosity (Bardgett et al., 2014). On the other hand, root penetration can also clog existing pores, and fine roots at low density may block pores and influence water movement (Huang et al., 2024). Coarse roots also decompose more slowly than fine roots. Importantly, root morphological traits significantly affect soil pore-size distribution and soil aggregates stability (Webb et al., 2022). Aggregate stability, a key indicator of soil structure (Six et al., 2000), is positively related to root length density (Rillig et al., 2002). The main reason is root length density can influence soil aggregate stability through physical enmeshment of soil particles by fine-root networks, chemical cementation by root exudates, and microbial stimulation leading to hyphal entanglement and the production of stabilizing compounds such as glomalin (Materechera et al., 1992; Morel et al., 1991; Rillig and Mummey, 2006). Hao et al. (2020) indicated that root length density indirectly influences soil infiltration by regulating soil aggregate stability and soil porosity. Despite the growing body of research on the effects of below-ground biomass on soil properties and soil infiltration rates (Freschet and Roumet, 2017; Huang et al., 2017), root morphological traits, which reflect plant adaptations for water acquisition, nutrient uptake, and drought tolerance in arid environments in arid environment, have received less attention regarding their direct and indirect regulatory roles in soil infiltration. Thus, the mechanisms by which fine-root traits modulate soil structure and infiltration remain poorly understood.

In the arid and semi-arid regions, where the primary objective is to control wind erosion and desertification, prioritizing shrub restoration over herbaceous plants is considered a more sustainable strategy for vegetation recovery, owing to shrubs' strong drought tolerance, short growth cycle, and high adaptability to climate fluctuations (Zhang et al., 2006; Zhou et al., 2023). Specifically, promoting the establishment of native shrubs has been identified as an effective approach to rehabilitate degraded and fragile soils (Bienes et al., 2016). As dominant components of arid and semiarid ecosystems, drought-tolerant shrub species provide essential ecosystem services, including soil stabilization, facilitation of herbaceous plant establishment, carbon sequestration, and restoration of wildlife habitats (Pallavicini et al., 2015). In water-scarce conditions, desert shrubs including *Caragana sinica*, *Artemisia sphaerocephala*, and *Zygophyllum xanthoxylum* can capture rainwater through stemflow and preferential flow pathways along roots, which benefits channel water to deeper soil layers and thereby reduces evaporative loss (Cammeraat et al., 2010). Moreover, sandy loam soils, which are common in arid regions, have low water-holding capacity and high infiltration rates, favoring the development of deep and extensive shrub root systems (Schenk and Jackson, 2002). Studies have indicated that shrubs with

higher root biomass positively influence soil porosity, water holding capacity and infiltration (Cai et al., 2021; Zhao et al., 2025). However, different shrub species exhibit distinct root morphological traits, leading to variations in their effects on soil porosity, aggregate stability, water-holding capacity and infiltration. Therefore, elucidating the root morphological traits of different shrub species is essential for providing a theoretical basis for species selection in arid land vegetation restoration.

Shrub vegetation significantly modifies soil porosity, water holding capacity and infiltration in arid and semi-arid regions (Xia et al., 2017). Deep soil water content governs the distribution of shrub species in arid ecosystem (Li et al., 2004), and enhanced soil infiltration facilitates the deep penetration of precipitation. This not only reduces surface evaporation losses but also increases deep soil water storage, providing a critical water source for deep-rooted plants during drought period (Li et al., 2004). Root morphological traits are key biological factors regulating soil structure, particularly in terms of water movement (Zadworny et al., 2018). However, most existing studies have focused either on the effects of vegetation establishment on soil properties and soil water dynamics or on the relationships between root traits and soil structure. Systematic research on how root traits affect hydrological function through soil structure remained scarce, limiting mechanistic understanding of vegetation-mediated hydrological processes in arid regions. Root effects on soil water infiltration differ markedly between root size classes. Coarse roots (> 2 mm) are known to contribute to preferential flow and macropore formation, particularly in deeper soil layers (Bi et al., 2025; Vinatier et al., 2024; Jacka et al., 2021). Fine roots (diameter ≤ 2 mm), in contrast, have rapid turnover and are unlikely to create stable, connected macropore networks. Instead, they mainly influence soil structure in the surface layer by promoting aggregate formation and stability through physical penetration but also promote aggregate formation and stability via physical entanglement of soil particles, chemical cementation by root exudates, and microbial stimulation that produces stabilizing compounds, thereby jointly regulating soil water movement. Thus, fine-root morphological traits primarily regulate surface soil infiltration by modifying soil structure, whereas coarse-root driven preferential flow dominates deeper water transport. Given this functional differentiation, unraveling the cascading mechanisms linking fine-root traits, surface soil structure, and infiltration function is essential for advancing both theoretical understanding and practical applications in arid land vegetation restoration.

Moreover, most previous studies on root-mediated infiltration have focused on root biomass or root length density as the primary predictors. Only a few have systematically compared multiple fine-root morphological traits, particularly in coarse-textured soils of arid regions. In this study, five fine-root (diameter ≤ 2 mm) morphological traits (i.e., specific root length, specific root surface area, root tissue density, root diameter and root length density) for three shrub species and a grass species in sandy loam soils. With this trait-based approach, we could distinguish between traits that hinder infiltration and those that promote it. That provides a more mechanistic understanding than has been available from biomass or root length density focused studies could achieve. The objectives of this study were to: (1) evaluate the role of fine-root morphological traits in regulating soil infiltration rates through their effects on soil structure and hydrological properties; (2) identify which shrub species are most conducive to vegetation restoration from the perspective of optimizing water infiltration, based on their fine-root morphological traits. We hypothesize that (1) In sandy loam soils, where macropores are the main pathways for water flow and the influence of soil matrix and swelling is limited, fine-root morphological traits regulate infiltration rates primarily by modifying soil pore structure. This hypothesis is specific to coarse-textured, low organic matter soils and is not intended to apply universally to all soil textures (2) Among the shrub species examined, those with fine-root morphological traits that enhance soil infiltration will be more conducive to vegetation restoration from the perspective of improving soil water infiltration. The findings of this study will provide a new sight into the contribution of

fine-root morphological traits to soil infiltration, informing the development of sustainable vegetation restoration strategies in arid regions.

2. Materials and methods

2.1. Study site

This experiment was conducted at an experimental site in Minqin County, Gansu province, China (103°1' E, 38°44' N). The site is located in the northeastern part of the Hexi Corridor at an elevation of 1307 m a.s.l. The region belongs to a temperate continental climate, with a mean annual temperature of 8.9 °C and a mean annual precipitation of 116.5 mm. Annual potential evapotranspiration is approximately 2350 mm.

The zonal natural vegetation in this region is typical desert steppe, dominated by *Calligonum mongolicum*, *Nitraria sphaerocarpa* and other drought-tolerant desert shrubs and herbs. The study area is a typical agro-pastoral ecotone suffering from serious land degradation, which is primarily driven by long-term strong wind erosion, extremely low precipitation, ultra-high potential evapotranspiration and intensive agricultural disturbance. The soil at the experimental site is classified as

sandy loam by the soil classification (WRB, 2014). The key physico-chemical properties in the 0–30 cm soil layer were determined before the experiment: available nitrogen 25.69 mg kg⁻¹, available phosphorus 15.53 mg kg⁻¹, available potassium 120.68 mg kg⁻¹, total nitrogen 7.04 g kg⁻¹, organic matter 12.19 g kg⁻¹, and soil pH 8.01 (Bai et al., 2023).

2.2. Experimental design

Three native desert shrub species (*Caragana sinica* (Buc'hoz) Rehder, *Artemisia sphaerocephala* Krasch., *Zygophyllum xanthoxylum* (Bunge) Maxim.) and one herbaceous species (*Astragalus laxmannii* Jacq.) were selected for this study. All species are typical native plants in the arid region and exhibit strong drought tolerance, meaning they can survive prolonged dry periods through deep rooting, reduced water loss, and physiological adjustments. The experiment was designed as a randomized complete block design, with four species and three replicates per species. Each species was assigned to a single monoculture plot per block. The plot size was 3.5 m × 3.5 m, and the distance between each plot was 1 m (Fig. 1). In June 2018, using hole sowing method, with three to five seeds per hole, followed by thinning to one healthy seedling

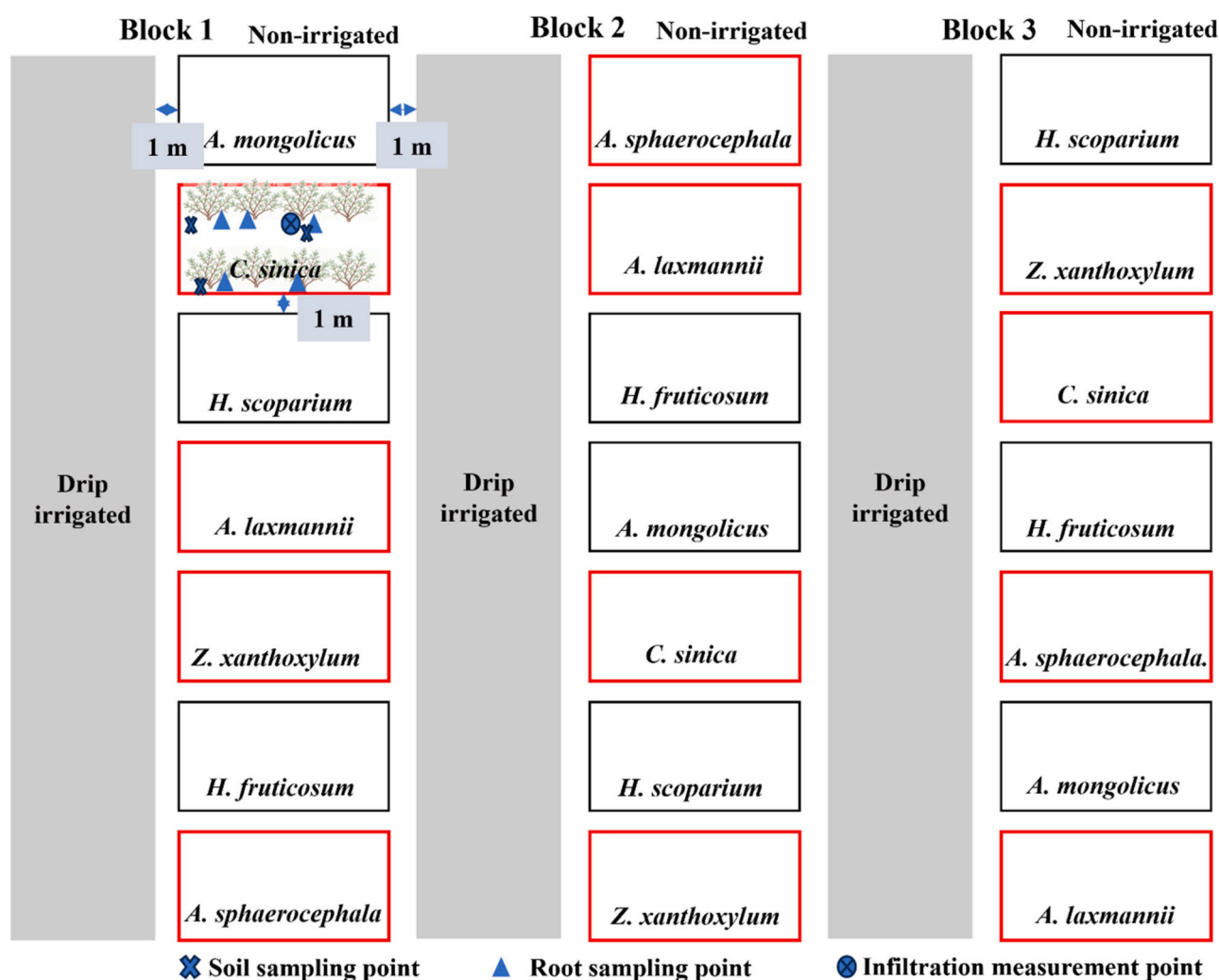


Fig. 1. Schematic diagram of the experimental design. The full experiment included seven plant species and two irrigation treatments (irrigated and non-irrigated) arranged in a randomised complete block design with three blocks. This study mainly concerns natural rainfall condition. Thus, only the non-irrigated treatments were selected. Each non-irrigated block contained 7 plots (3.5 m × 3.5 m). The red bold-bordered plots indicate the 12 plots used in this study. Within each of these 12 plots, a single infiltration measurement was performed at the centre, and five individual plants were selected for root sampling (0–20 cm, 20–40 cm) and separate set of three plants was selected for soil sampling (0–20 cm). The replicate samples within each plot were averaged to a single plot-level value, with the plot as the independent experimental unit.

per hole after emergence. For *C. sinica* and *A. sphaerocephala* seeds were sown at a spacing of 1 m between rows and between plants, resulting in 16 plants per plot. For *Z. xanthoxylum* and *A. laxmannii*, the spacing was 0.5 m, giving 64 plants per plot. Prior to the initiation of the present experiment, the experimental site had been under continuous cultivation of silage maize (*Zea mays* L.) for five consecutive years. No irrigation or fertilization was applied during the experimental period, allowing the plants to grow under natural precipitation conditions. Vegetation establishment was successful, with all species surviving and showing normal growth by the end of the first growing season. The plots remained undisturbed and unchanged from 2018 to 2023. Soil samples were collected in July 2023 during the peak growing season.

2.3. Sampling and measurement procedures

In each plot, five individual plants were randomly selected. Root samples were collected near the base of study plants using a 9-cm diameter auger in 0–20 cm and 20–40 cm soil depth. Preliminary examination showed that the 0–20 cm layer contained very few fine roots (diameter ≤ 2 mm), whereas the 20–40 cm layer contained the majority of fine roots. Therefore, fine-root trait analysis was based on samples from the 20–40 cm depth. Root samples were taken back to laboratory and carefully washed with water to remove adhering soil. After washing, the entire root system was spread on a white tray. Fine roots (diameter ≤ 2 mm) were manually separated from coarse roots using fine forceps and a digital caliper. The separated fine roots were then scanned at a resolution of 300 dpi by using an Epson Expression 10000XL scanner (EPSON America Inc.). The scanned image was analyzed with the WinRhizo 2019pro root system image analysis system (Regent Instruments Inc., Canada, since 1991) to obtain mean root diameter (RD ; cm), root volume (RV ; cm³), root surface area (SA ; cm²), and total root length (RL ; m). After scanning, the roots were oven-dried at 80 °C for 48 h until constant weight to determine root dry weight (RDW ; g). Fine-root morphological traits were calculated as follows:

$$RTD = RDW/RV \quad (1)$$

$$SRL = RL/RDW \quad (2)$$

$$RSA = SA/RDW \quad (3)$$

where RTD is the root tissue density (g cm⁻³), SRL is the specific root length (m g⁻¹), and RSA is the specific root surface area (cm² g⁻¹).

Soil bulk density (BD ; g cm⁻³) and field capacity (FC ; %) were measured on undisturbed soil cores collected from the 0–20 cm soil depth as a single layer using the cutting ring (5-cm diameter and 5-cm height). FC was determined using the gravitational drainage method. Specifically, undisturbed soil cores were saturated by submerging in water for 48 h. The cores were then placed on a layer of air-dried soil to drain freely under gravity. After 2 h of drainage, the cores were weighed (M_1). They were then left to drain for an additional 6 h and weighed once more (M_2). Finally, the cores were oven-dried at 105 °C for 72 h to constant weight (M_3). Capillary holding capacity (CHC) and FC was calculated as:

$$CHC = (M_1 - M_3)/(M_3 - M_0) \quad (4)$$

$$FC = (M_2 - M_3)/(M_3 - M_0) \quad (5)$$

where M_0 is the weight of the empty cutting ring (g).

Soil water-stable aggregates refer to the fraction of soil aggregates that remain intact after being subjected to wetting and disruptive forces. They were determined using a TPF-100 soil aggregate analyzer. Undisturbed soil samples were excavated from the 0–20 cm soil depth and placed in aluminum boxes. In the laboratory, each soil block was gently hand-broken along natural fissures into small clumps of 1–2 cm in diameter and then air-dried. Air-dried subsamples (50 g each) were

soaked for 30 min, then placed on a nest of sieves with apertures of 5, 3, 2, 1, 0.5, and 0.25 mm. The sieves were then oscillated vertically in water at 30 cycles per minute for 3 min. The aggregates retained on each sieve were collected, oven-dried at 60 °C to constant weight, and weighed. The calculation of soil porosity characteristics (Jiao et al., 2011) and mean weight diameter (Parent et al., 2012) were as follows:

$$TP = (1 - BD/ds) \times 100\% \quad (6)$$

$$CP = (CHC \times BD)/\rho \times 100\% \quad (7)$$

$$NCP = TP - CP \quad (8)$$

$$MWD = \sum_{i=1}^N \bar{X}_i W_i \quad i = (1, 2, \dots, n) \quad (9)$$

where TP is the total porosity (%), CP is the capillary porosity (%), NCP is the non-capillary porosity (%), ds is the soil particle density, in this study ds is 2.65 g cm⁻³, ρ is water density and assuming 1 g cm⁻³, MWD is mean weight diameter, $\bar{X}_i$ is the assumed diameter for i th fraction, W_i is the weight fraction remained on the i th sieve-size. N is the number of sieves, $N = 6$.

Soil infiltration rate was determined using the double-ring infiltrometer method. The apparatus consisted of an inner ring (20 cm diameter) and an outer ring (35 cm diameter), both 25 cm in height. For each plant treatment, three replicate measurements were performed on a flat surface near the plant's tap-root zone. Surface litter was cleared before the rings were gently driven approximately 8 cm vertically into the soil. A thin plastic sheet was placed on the soil surface inside both rings to minimize disturbance during water addition. Water was poured onto the sheet, which was then removed once the target water level (5 cm) was reached. At the start of each test, water was poured simultaneously into both rings to a depth of 5 cm, and the stopwatch was started immediately. The time required for the water level to drop by 5 mm was recorded. Whenever the water level fell below 4 cm, it was replenished to 5 cm, ensuring that the water levels in the inner and outer rings remained equal throughout the experiment. The measurement was stopped once steady-state infiltration was achieved, specifically when the time required for the water level to drop by 5 mm became nearly constant over three successive measurements (the coefficient of variation of the three time readings was less than 5%) and the total measure timing over than 1 h. Prior to the infiltration tests, the initial soil water content (SWC , 0–20 cm) was determined gravimetrically. No significant differences were detected among the four plant treatments ($P > 0.05$, Table 1). The infiltration rate was calculated as follow:

$$i = \Delta I / \Delta t \quad (10)$$

where i is the infiltration rate (mm min⁻¹), ΔI is the cumulative infiltration amount in the inner ring (mm) during Δt time of infiltration

Table 1

Soil physical properties at 0–20 cm soil depth under different shrub species and grass in 2023.

	FC (%)	TP (%)	CP (%)	NCP (%)	MWD (mm)	SWC (%)
<i>A. laxmannii</i>	19.0 ± 1.6	46.4 ± 2.7	32.8 ± 1.8	13.6 ± 1.0ab	0.47 ± 0.0b	2.6 ± 0.3a
<i>C. sinica</i>	17.1 ± 2.1	44.4 ± 1.9	31.8 ± 3.2	12.6 ± 1.5b	0.21 ± 0.0c	2.8 ± 0.2a
<i>A. sphaerocephala</i>	17.2 ± 2.6	47.0 ± 2.5	29.9 ± 2.7	17.2 ± 1.4a	0.65 ± 0.1a	2.7 ± 0.1a
<i>Z. xanthoxylum</i>	19.5 ± 1.7	47.3 ± 3.9	33.6 ± 2.3	13.7 ± 3.7ab	0.64 ± 0.0a	2.7 ± 0.2a

Notes: FC: Field capacity; TP: Total porosity; CP: Capillary porosity; NCP: Non-capillary porosity; MWD: Mean weight diameter of water-stable aggregates; SWC: Soil water content. Different lowercase letters indicate statistically significant differences among the species ($p < 0.05$).

(min).
Based on the infiltration curve, the initial infiltration rate (IR) was defined as the average rate during the first 3 min, the steady infiltration rate (SIR) as the average rate over the last 3 min, and the average infiltration rate (AIR) as the mean rate over the entire measurement period.

2.4. Statistical analysis

One-way analysis of variance (ANOVA) followed by Tukey’s HSD post-hoc test was used to examine the significant differences in soil properties and fine-root morphological traits among the different planting species at 0.05 level. Pearson correlation analysis was used to analyze the relationship between fine-root morphological traits and soil properties. No correction for multiple comparisons was applied to the correlation analyses, as they were exploratory and involved a limited number of a priori specified pairs. Exact p-values are reported, and results should be interpreted with caution. Stepwise multiple regression was performed to identify the main factors among fine-root morphological traits and soil properties influencing IR and SIR. Given the small sample size (n = 12), the stepwise regression results should be interpreted with caution, as high R² values may partly reflect overfitting. The findings are exploratory and require validation with larger datasets. To elucidate the direct and indirect effects of fine-root morphological traits on IR through soil structure properties and water-holding capacity, a structural equation model (SEM) was applied. Given the small sample size (n = 12), the SEM was used as an exploratory tool for hypothesis generation rather than confirmatory testing. All statistical analyses, except for the SEM, were performed using SPSS Statistics 25.0 (IBM, USA). All figures were created in SigmaPlot 14.0 ((Systat Software, Inc., USA). The SEM was fitted and evaluated using R software (version 4.3.3) with the *sem* package.

3. Results

3.1. Soil properties of grass and shrubs

NCP of *A. sphaerocephala* was the highest among the four species, but it was significantly higher only than that of *C. sinica* (p < 0.05). Both *A. sphaerocephala* and *Z. xanthoxylum* formed soils with significantly greater MWD of water-stable aggregates than *C. sinica* (p < 0.001) and *A. laxmannii* (p = 0.001).

The infiltration rates during the whole infiltration process of *A. laxmannii* and *Z. xanthoxylum* were lower than that of *C. sinica* and *A. sphaerocephala* (Fig. 2A). IR of *C. sinica* was 30% and 50% higher than

that of *A. laxmannii* and *Z. xanthoxylum* (Fig. 2B). Moreover, the difference of SIR between *C. sinica* and *A. sphaerocephala* was not significant, but significantly higher than that of *A. laxmannii* and *Z. xanthoxylum* (P < 0.05).

3.2. Fine-root traits of different plant species

The SRL and SRSA of *A. laxmannii*, *C. sinica* and *A. sphaerocephala* significantly higher than that of *Z. xanthoxylum* (p < 0.05, Table 2). *C. sinica* exhibited the highest RD, which was significantly greater than that of *A. sphaerocephala* and *Z. xanthoxylum* by 45% (p < 0.05). In contrast, *A. laxmannii* had the lowest RD, which was significantly lower than that of *C. sinica* by approximately 47% (p < 0.05). *A. sphaerocephala* had the lowest RLD, which was significantly lower than that of *C. sinica* by approximately 94% (p < 0.05) and lower than those of *A. laxmannii* and *Z. xanthoxylum* by 89%. Overall, among the four plant species, *A. laxmannii* had the highest SRSA, *A. sphaerocephala* showed higher SRL and RTD, *C. sinica* had higher RD and RLD, and *Z. xanthoxylum* performed relatively well in RTD (Fig. 3).

3.3. Relationship between fine-root traits and soil properties

Given the small sample size (n = 12), all reported correlations are considered exploratory and are presented as associations rather than definitive relationships. SRSA and RD showed positive associations with IR, SIR and AIR (p < 0.05). Conversely, RTD exhibited a negative association with infiltration rates (p < 0.01). SIR and AIR were negatively associated with FC (p < 0.05), FC was negatively associated with SRSA

Table 2
Fine-root morphological traits at 20–40 cm soil depth under different shrub species and grass in 2023.

Species	SRL (m g ⁻¹)	SRSA (cm ² g ⁻¹)	RTD (g cm ⁻³)	RD (mm)	RLD (cm cm ⁻³)
<i>A. laxmannii</i>	57.4 ± 1.8a	471.3 ± 19.8ab	0.2 ± 0.0	0.3 ± 0.0c	0.2 ± 0.1ab
<i>C. sinica</i>	34.3 ± 23.2ab	469.5 ± 251.6ab	0.1 ± 0.0	0.6 ± 0.0a	0.3 ± 0.2a
<i>A. sphaerocephala</i>	51.7 ± 10.8a	557.4 ± 73.2a	0.1 ± 0.0	0.4 ± 0.0b	0.1 ± 0.0b
<i>Z. xanthoxylum</i>	22.0 ± 3.5b	271.2 ± 44.1b	0.2 ± 0.1	0.4 ± 0.1b	0.2 ± 0.1ab

Notes: SRL: specific root length; SRSA: Specific root surface area; RTD: root tissue density; RD: root diameter; RLD: root length density. Different lowercase letters indicate statistically significant differences among the species (p < 0.05).

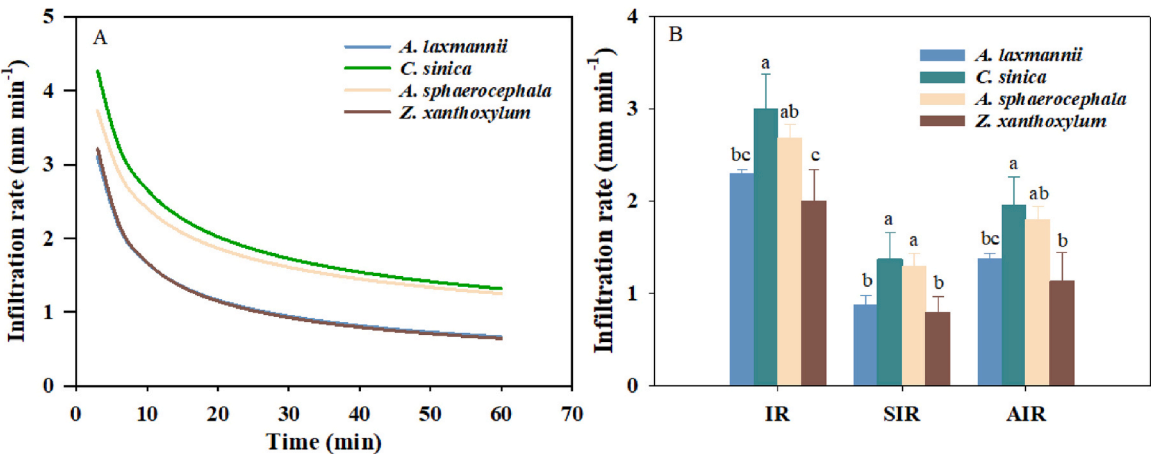


Fig. 2. Changes of soil infiltration rate along with time (A) and initial infiltration rate (IR), steady infiltration rate (SIR) and average infiltration rate (B, AIR) of *A. laxmannii*, *C. sinica*, *A. sphaerocephala* and *Z. xanthoxylum* in 2023. Different lowercase letters indicate statistically significant differences among the species (p < 0.05).

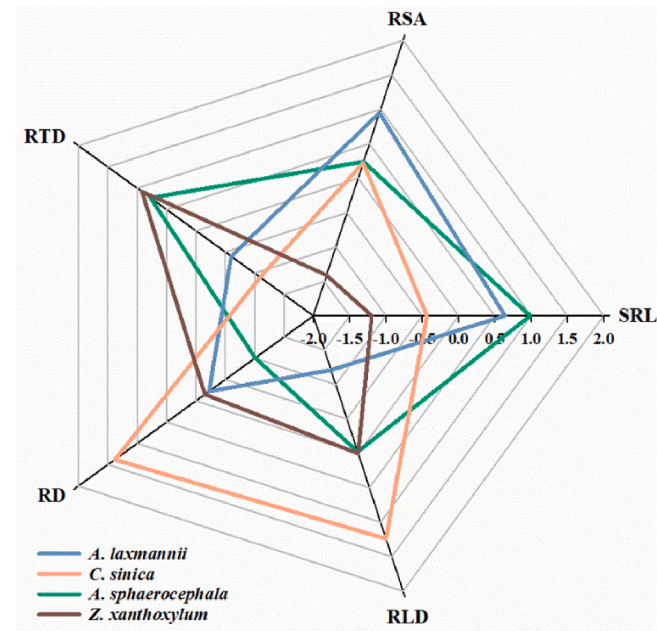


Fig. 3. The comprehensive performance of specific root length (SRL), specific root surface area (SRSA), root tissue density (RTD), root diameter (RD), root length density (RLD) of *A. laxmannii*, *C. sinica*, *A. sphaerocephala* and *Z. xanthoxylum* at 20–40 cm soil depth in 2023. All data were standardized.

($p < 0.05$), whereas positively associated with TP and CP ($p < 0.05$). Moreover, TP was positively associated with RTD ($p < 0.05$), CP showed a negative association with SRL and RSA ($p < 0.05$), NCP was negatively correlated with RLD ($p < 0.05$). MWD was negatively associated with

RD, RLD and IR ($p < 0.05$) (Fig. 4).

3.4. The effects of fine-root traits and soil properties on soil water infiltration

For IR, the model identified RTD, SRSA, RLD and NCP were the main affect factors (Table 3). RTD exerted the strongest negative effect ($\beta = -6.03$), whereas SRSA, RLD and NCP exhibited positive contributions. SIR was similarly well-explained ($R^2 = 0.93$) by RTD, NCP, and MWD. RTD displayed a pronounced negative influence ($\beta = -3.957$), while NCP positively affected SIR. Notably, MWD also negatively correlated with SIR.

The SEM showed excellent fit to the data (Fig. 5, $\chi^2 = 0.044$, $df = 2$, $p = 0.978$; CFI = 1.000; RMSEA = 0.000; SRMR = 0.014; GFI = 0.998). RTD exhibited the strongest negative influence, with a direct effect of -0.74 and total effect of -0.80 . MWD showed a moderate negative direct effect that was slightly attenuated in the total effect. TP demonstrated a notable positive direct effect that decreased in the total effect. FC maintained consistent negative direct and total effects, underscoring its stable role in water retention that uniformly reduces infiltration capacity.

4. Discussion

Vegetation restoration drives root-soil interactions that modify soil hydrological properties, particularly infiltration, with fine-root morphological traits playing differential roles in mediating these changes. While many studies report a negative effect of RLD on soil infiltration rate (Liu et al., 2020; Hao et al., 2020), our results showed only a weak relationship between RLD and infiltration, with RTD emerging as a more influential trait. Specifically, RTD exerted a significant negative effect on soil infiltration rate. Higher RTD associated with

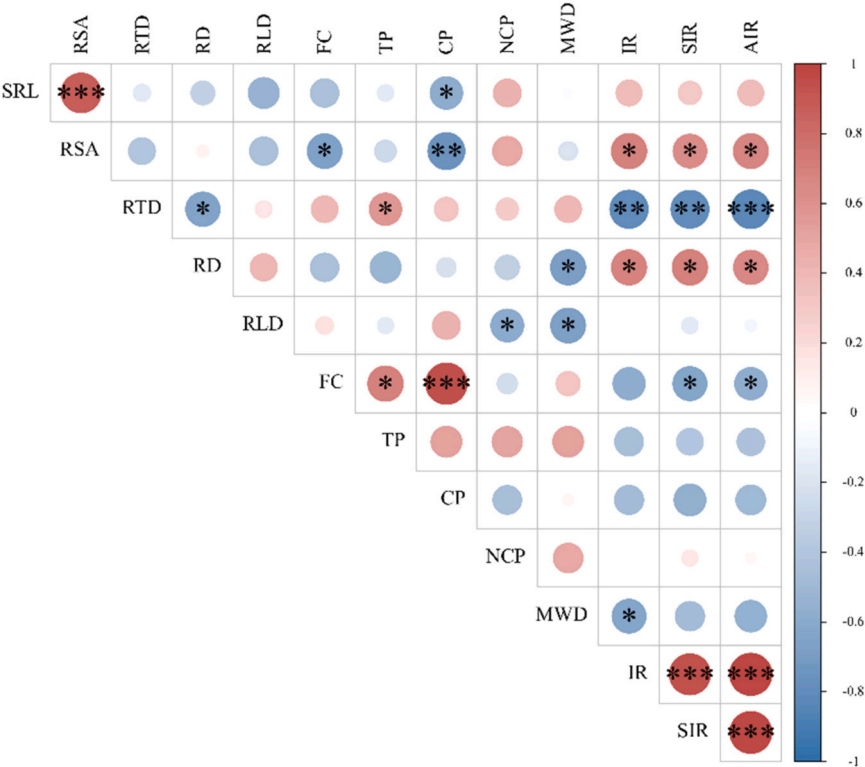


Fig. 4. The correlation analysis between fine-root morphological traits (20–40 cm) and soil properties (0–20 cm). SRL, SRSA, RTD, RD and RLD are specific root length, specific root surface area, root tissue density, root diameter, root length density, respectively. FC is field capacity. TP, CP and NCP are total porosity, capillary porosity and non-capillary porosity. MWD is the mean weight diameter of water-stable aggregates. IR, SIR and AIR are the initial, steady and average infiltration rate. Note: * refers to $P < 0.05$; ** refers to $P < 0.01$; *** refers to $P < 0.001$.

Table 3
Stepwise regression models predicting initial infiltration rate (IR) and steady infiltration rate (SIR) based on fine-root traits and soil physical properties.

Stepwise regression equation	R ²	P	VIF
IR= 1.75–6.03RTD + 0.001SRSA + 1.79RLD + 0.06NCP	0.97	< 0.001	1.91–3.01
SIR= 1.10–3.96RTD + 0.07NCP–0.74MWD	0.93	< 0.001	1.20–1.46

Notes: RTD: root tissue density; SRSA: specific root surface area; RLD: root length density; NCP: Non-capillary porosity; MWD: Mean weight diameter of water-stable aggregates.

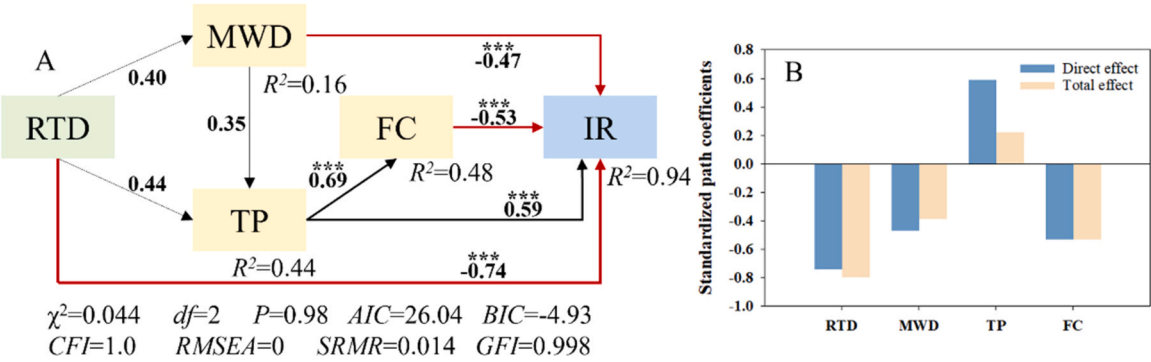


Fig. 5. The path analysis represented the hypothesized casual relationships of root tissue density (RTD) on initial infiltration rate (IR) through total porosity (TP), mean weight diameter of water-stable aggregates (MWD) and field capacity (A, FC). B showed the direct and indirect standardized path coefficients of RTD, MWD, TP and FC. The thickness of the arrow line indicated the strength of the relationship. Black and red arrows indicated the effects was positive and negative, respectively. *** refers to $P < 0.001$.

increased CP and field capacity FC, collectively increasing soil water holding capacity but impede water movement. This conclusion was supported by our structural equation model, which identified RTD having the strongest direct negative effect on infiltration rate, followed by significant negative effects from MWD and FC. Nevertheless, in this study, FC was measured with a gravitational drainage method. This method only provides water content after short-term drainage. This did not give the full retention curve (e.g., wilting point, plant-available water), and lacks the measurement of water potential, which limits comparability with studies that define FC at a specific matric potential. Therefore, our interpretations of “water retention” are restricted to the specific drainage conditions of this study. Future studies should consider using pressure plate or dew-point methods to determine permanent wilting point and plant-available water.

The negative effect of high RTD on soil infiltration rate can be attributed to a confluence of physical and biological factors. First, roots with higher RTD generally have longer lifespans and lower turnover rate (Eissenstat et al., 2000; Kramer-Walter et al., 2016), which reduces the frequency of root decay and thus limits the formation of persistent root channels. Second, high RTD is often linked to greater contents of hydrophobic compounds such as waxes and lignin, which can induce soil water repellency and further restrict water infiltration (Hallett et al., 2003). The net result of these intertwined mechanisms is a shift in pore size distribution towards a dominance of smaller pores. While this shift enhances water retention (reflected in higher FC), it concurrently imposes a significant constraint on soil infiltration rate.

Conversely, higher RD and RSA were positively related to soil infiltration rate. This can be primarily attributed to the distinct roles that these traits play in creation and modification of soil pore architecture. First, a larger RD, particularly within the fine-root spectrum (0–2 mm in this study), enhances the capacity to create and stabilize larger biopores during root growth and, importantly, following root decay. Thicker roots leave behind more persistent channel-like voids upon decomposition, which are conducive to enhancing preferential flow paths (Bardgett et al., 2014). In the present study, RD within the range of 0–2 mm, within this range, a larger fine-root diameter correlates with the formation of larger biopores following root decomposition, which is more conducive to water movement (Cui et al., 2019). Moreover, higher

RSA indicated the plant with abundant fine roots. Upon senescence and decay, this network of fine roots creates an extensive and interconnected root channels (Ghestem et al., 2011). This enhanced connectivity among pores significantly reduces the tortuosity of water flow paths, thereby increasing the overall hydraulic conductivity and infiltration capacity of the soil.

Recent work has made it clear that preferential flow through root can strongly shape how water moves into the soil. Jacka et al. (2021) found that different tree species alter infiltration and preferential flow mainly by changing pore size distribution and connectivity. Vinatier et al. (2024) showed that quasi-steady infiltration under ponded conditions is not fixed; it varies with plant cover and management, which points to the dynamic nature of root-created pathways. In arid regions, Bi et al. (2025) observed that arbuscular mycorrhizal fungi can boost root growth and enhance preferential flow in coal mine wastes, underlining the real-world importance of root-associated macropores for water transport. Our own study looked only at fine roots (≤ 2 mm) in the surface layer, so we likely missed what coarse roots (> 2 mm) and their decay channels do in deeper soil. That part of preferential flow is not captured here. Future work should look at the whole root system, across all diameters and depths, to understand how root-created flow paths affect infiltration in arid shrublands.

In our study, soil properties (bulk density, porosity, field capacity, aggregate stability) were measured in the 0–20 cm soil layer, whereas fine-root traits were measured at 20–40 cm depth. We reserved the 0–20 cm layer for undisturbed soil property measurements (critical for infiltration characterization) and sampled roots from the 20–40 cm layer where fine roots of the studied shrubs were most abundant and where sampling would not disturb the surface soil. However, we acknowledge that infiltration processes are primarily controlled by near-surface soil structure (0–20 cm). Therefore, linking deeper root traits to surface soil properties may not fully capture the direct influence of roots present in the surface layer. Ideally, root traits and soil properties should be measured in the same depth interval. Future studies should consider sampling roots from the 0–20 cm layer (e.g., using smaller-diameter augers to minimize disturbance) to allow a more direct comparison.

Notably, the results showed that RD had a negative relationship with

aggregate stability. The aggregates in the sandy soil are very fragile, and the interweaving of fine root systems may cause damage to them (Angers and Caron, 1998). Moreover, contrary to the study in finer-textured soils (Hao et al., 2020), in this study we have found that aggregate stability had negative effects on soil infiltration rate. We propose that this distinct can be explained by the unique context of sandy loam soils. In such coarse-textured soil, highly stable aggregates might primarily consist of finer particles that clog macropores or reduce connectivity between pores, also increase water-holding capacity, thereby impeding preferential flow paths crucial for infiltration (Assouline, 2004). Moreover, sandy soil with lower organic matters that result the aggregate structure is prone to disintegrate when exposed to water, thereby reducing the infiltration rate (Lado et al., 2004).

Our findings reveal distinct functional strategies among root system in modifying soil properties. For different shrub species, they each have their own strengths in terms of certain morphological traits. *A. sphaerocephala* had the higher RSA and SRL. Soil infiltration rate of *A. sphaerocephala* was second only to the *C. sinica*, while with the highest water-stable aggregates. It demonstrated a remarkable ability to balance infiltration enhancement with soil stabilization, exhibiting the second-highest infiltration rate while maintaining the most stable soil aggregates. This dual capacity suggests its potential for simultaneously improving groundwater recharge and erosion control in sandy ecosystems. The high RSA and SRL further enhance its competitive advantage in nutrient and water acquisition from coarse-textured soils (Tan et al., 2023). This combination of traits positions *A. sphaerocephala* as a keystone species for ecological restoration in water-limited environments.

Furthermore, *A. laxmannii* and *Z. xanthoxylum* exhibited a higher water retention capacity through increasing CP and FC, albeit at the expense of reduced infiltration rates. This water conservation strategy, while limiting deep percolation, may create favorable microsites for seedling establishment by maintaining higher surface soil moisture (Guerrero-Campo et al., 2006). High RTD represents a key adaptation to arid and infertile conditions, increasing root lifespan to improve resource retention capacity while enhancing resistance to soil mechanical pressure and wind erosion (Kramer-Walter et al., 2016).

Thus, based on the fine-root traits and their effects on soil properties, especially the significance of soil infiltration rate on arid environments, we propose that the combination of *A. sphaerocephala* with either *A. laxmannii* or *Z. xanthoxylum* may contribute to multiple soil hydrological functions in arid land restoration. Species with contrasting ecological niches can enhance overall ecosystem performance through resource partitioning and facilitation (Chesson et al., 2000). Specifically, *A. sphaerocephala* showed higher infiltration rate, whereas *A. laxmannii* and *Z. xanthoxylum* exhibited higher water retention at field capacity. However, whether this complementarity would translate into improved deep infiltration, surface water retention, and overall vegetation community resilience under field conditions remains to be tested. Future studies with mixed-species plantings, longer monitoring periods, and measurements of coarse-root and preferential flow processes are needed to validate these hypotheses.

5. Conclusion

This study suggests how fine-root morphological traits are associated with soil water infiltration in arid areas. In sandy loam soil, RTD showed a direct negative effect on the infiltration rate by increasing soil CP and FC, whereas RD and RSA significantly promote water infiltration by creating and maintaining steady biopores. Specially, higher aggregate stability was negatively associated with water infiltration in the sandy soil. The four plant species exhibited complementary ecological strategies through divergent fine-root morphological traits. *A. sphaerocephala* showed a combination of higher infiltration rates and higher aggregate stability, whereas *A. laxmannii* and *Z. xanthoxylum* showed higher water retention at field capacity. These findings address the two study

objectives: (1) fine-root traits are differentially associated with infiltration through influence soil structure and water retention; (2) based on their root traits, *A. sphaerocephala* appears promising for enhancing infiltration, while *A. laxmannii* and *Z. xanthoxylum* may favour surface water retention. Therefore, under the specific conditions of this study, combining *A. sphaerocephala* with either *A. laxmannii* or *Z. xanthoxylum* may help achieve complementary effects on water infiltration and surface water preservation. However, these findings are based on a limited number of species and a single site, whether such combinations would enhance plant community adaptability and ecosystem sustainability in the long term requires further investigation under field conditions. Moreover, soil water-holding capacity was limited to CHC and FC without water potential measurement, which restricts comparability with pressure-plate-based studies.

CRediT authorship contribution statement

Xiaowen Hu: Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Yinan Chen:** Investigation, Data curation. **Dongfeng Ma:** Investigation. **Mengjie Bai:** Investigation, Data curation. **Yue Zhang:** Investigation, Data curation. **Ze Huang:** Writing – review & editing, Writing – original draft, Visualization, Investigation, Funding acquisition, Formal analysis, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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